

Program – 10

AIM : To write a Java program to produce tokens from a given string.

CODE :

```
import java.util.*;

public class StringTokens {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");

        String str = sc.nextLine();

        // Using StringTokenizer

        StringTokenizer st = new StringTokenizer(str);

        System.out.println("Tokens are:");

        while (st.hasMoreTokens()) {

            System.out.println(st.nextToken());

        }

        sc.close();

    }

}
```

OUTPUT :

```
Enter a string: Hello world welcome to Java
Tokens are:
Hello
world
welcome
to
Java

=== Code Execution Successful ===
```

Program – 11

AIM : To write a Java program to check whether two strings are anagram or not.

CODE :

```
import java.util.*;

public class AnagramCheck {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first string: ");

        String str1 = sc.nextLine();

        System.out.print("Enter second string: ");

        String str2 = sc.nextLine();

        // Convert to lowercase and remove spaces
        str1 = str1.replaceAll("\\s", "").toLowerCase();
        str2 = str2.replaceAll("\\s", "").toLowerCase();

        char[] arr1 = str1.toCharArray();
        char[] arr2 = str2.toCharArray();

        Arrays.sort(arr1);
        Arrays.sort(arr2);

        if (Arrays.equals(arr1, arr2)) {
```

```
        System.out.println("Strings are Anagram");  
    } else {  
        System.out.println("Strings are NOT Anagram");  
    }  
  
    sc.close();  
}  
}
```

OUTPUT :

```
Enter first string: listen  
Enter second string: slient  
Strings are Anagram  
  
=== Code Execution Successful ===
```

```
Enter first string: dev  
Enter second string: saini  
Strings are NOT Anagram  
  
=== Code Execution Successful ===
```

Program – 12

AIM : To write a Java program to find the percentage of uppercase, lowercase, digits, and special characters in a string.

CODE :

```
import java.util.*;

public class CharacterPercentage {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        int upper = 0, lower = 0, digit = 0, special = 0;

        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);

            if (Character.isUpperCase(ch))
                upper++;
            else if (Character.isLowerCase(ch))
                lower++;
            else if (Character.isDigit(ch))
                digit++;
            else
                special++;
        }
    }
}
```

```
}

int total = str.length();

System.out.println("Uppercase Percentage: " + (upper * 100.0 / total));
System.out.println("Lowercase Percentage: " + (lower * 100.0 / total));
System.out.println("Digits Percentage: " + (digit * 100.0 / total));
System.out.println("Special Characters Percentage: " + (special * 100.0 / total));

sc.close();
}
}
```

OUTPUT :

```
Enter a string: Hello@123
Uppercase Percentage: 11.11111111111111
Lowercase Percentage: 44.44444444444444
Digits Percentage: 33.333333333333336
Special Characters Percentage: 11.11111111111111

=== Code Execution Successful ===
```

Program – 13

AIM : To write a Java program to reverse and toggle each word in a string.

CODE :

```
import java.util.*;

public class ReverseToggleWords {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");

        String str = sc.nextLine();

        String[] words = str.split(" ");

        String result = "";

        for (String word : words) {

            String reversed = "";

            // Reverse word

            for (int i = word.length() - 1; i >= 0; i--) {

                char ch = word.charAt(i);

                // Toggle case

                if (Character.isUpperCase(ch))

                    reversed += Character.toLowerCase(ch);

            }

        }

        System.out.println(result);
    }
}
```

```
        else
            reversed += Character.toUpperCase(ch);
        }

        result += reversed + " ";
    }

    System.out.println("Result: " + result.trim());

    sc.close();
}
}
```

OUTPUT :

```
Enter a string: Hello World
Result: OLLEh DLR0w

=== Code Execution Successful ===
```